A 47-nW Voice Activity Detector (VAD) Featuring a Short-Time CNN Feature Extractor and an RNN-Based Classifier With a Non-Volatile CAP-ROM

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***Abstract*— This article reports an area-and-power-efficient voice activity detector (VAD) for voice-control edge devices. It innovates a short-time convolutional neural network (ST-CNN) and a recurrent neural network (RNN)-based classifier. Such a classifier shortens the extraction window of the ST-CNN while reducing its signal leakage, detection latency, and area and power budgets. The RNN also aids in parameter reduction of the VAD to only 45. We also propose the non-volatile capacitor-ROM (CAP-ROM) as the weight storage, eliminating the volatile memory and related memory access while freeing the VAD from the weight preloading procedure before activation. The non-reconfigurability of the CAP-ROM is acceptable since we verify that the VAD does not strongly depend on the dataset. Training with the Google speech command dataset (GSCD), our VAD in 65-nm CMOS exhibits a 94%/91% overall hit rate on the GSCD/TIMIT dataset with small power (47 nW) and area (0.022 mm^2). There is no significant degradation of the hit rate for the supply voltage from 0.9 to 1.3 V or temperature from 0 to 60 °C, substantiating the robustness of the VAD.**

***Index Terms*— Capacitor-rom (CAP-ROM), edge computing, feature extraction, model reduction, non-volatile memory, receptive field, recurrent neural network (RNN), switched-capacitor circuits, voice activity detector (VAD).**

I. Introduction

SPEECH recognition, as an important interface for enabling various natural language processing tasks, attracts increasing attention due to the rapid development of deep neural networks [1]. Speech recognition is computation and memory intensive with a low activity rate [2]. Thus, an always-on voice activity detector (VAD) [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13] is a promising method to save power by deactivating speech recognition when absent of a human voice. Designing an energy-and-area-efficient VAD for adopting the battery with limited area and system integration is crucial. Moreover, a VAD should guarantee a hit rate with low latency. Although a traditional microphone consumes much larger power than a VAD, a low-power (32 nW) ON-chip MEMS microphone in [6] still motivates the low-power VAD design.

A VAD generally consists of a feature extractor [14], [15], [16] and a classifier [17], [18], [19]. The classifier evaluates a VAD output using the extracted features with an extraction window length (t_{window}) of tens of milliseconds. A digital VAD [3] was insensitive to circuit noise and mismatch but consumed considerable power (>20 μW). Feature extractors based on the analog filter bank [4], [5] avoided the full bandwidth analog-to-digital converter and a digital feature extractor. It was power-and-area consuming to process the low-frequency signal with multiple channels in parallel. A mixer-based analog feature extractor [6] reduced its power and area by scanning the frequencies in a time-interleaved manner, but it drastically increased the latency to 512 ms. A time-domain convolutional neural network (TD-CNN)-based feature extractor [7] stored the input signal in the analog memory and extracted features with a switched-capacitor circuit which avoids the power-and-area-hungry analog filter bank. However, even though t_{window} decreased to 10 ms in the TD-CNN, a large analog memory was necessary to reduce the signal leakage. Conventional VADs [4], [5], [6], [7] used memoryless classifiers, leading to long t_{window}. Moreover, a neural network-based classifier requires ON-chip volatile memory to store the weight. The SRAM [6] or weight register [7] usually consumes significant power and area of a VAD and requires weight preloading before activating the VAD.

Inspired by digital signal processing, the finite impulse response (FIR) filter only has a feed-forward path, while the infinite impulse response (IIR) filter has recurrent memory. The latter can obtain similar rejection ability with lower orders and thus fewer parameters than its counterparts. According to the analogy to the FIR filter, a memoryless classifier [Fig. 1(a)] evaluates a result with the features within an extraction window.

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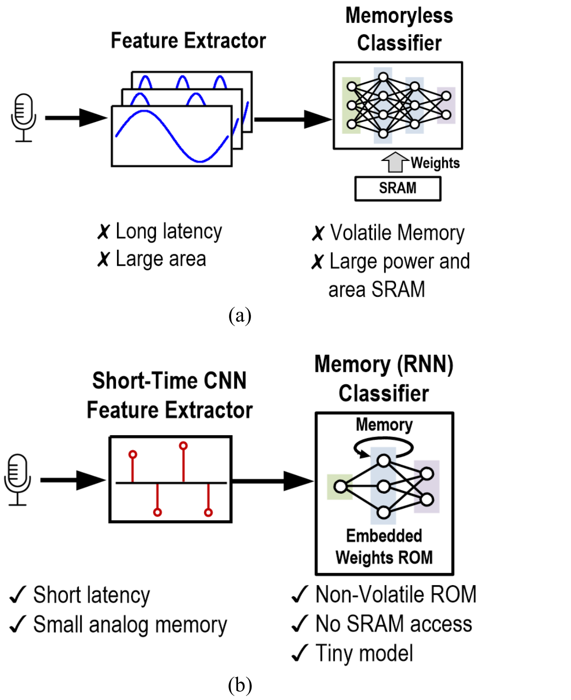

Fig. 1. (a) Conventional memoryless VAD classifier. (b) Proposed memory VAD with an ST-CNN.

Therefore, we propose to deploy a recurrent neural network (RNN)-based classifier [Fig. 1(b)] that evaluates a result with the current features and the past information from the recurrent memory, as presented in [20]. Giraldo et al. [21] used a memory classifier to extend the extraction window for keyword spotting and speaker verification. Here, we use an RNN-based classifier to reduce t_{window} to deploy a short-time-CNN (ST-CNN) which reduces the area of the analog memory while having less signal leakage. Also, a smaller t_{window} reduces the VAD latency to 2.5 ms and the model size to 45 parameters. Finally, we quantize the output of the ST-CNN and RNN-based classifier to 1 bit, leading to a discretized output distribution that desensitizes the comparator to the offset and noise.

For storing the weight, we propose a capacitor-ROM (CAP-ROM) that uses the capacitor as a non-volatile memory. Thus, we can eliminate the ON-chip SRAM and its corresponding memory access. We can also avoid the weight preloading process before activating the VAD. The capacitor can take a continuous value, only limited by the design rules. It avoids weight quantization which usually leads to hit rate degradation. Using the CAP-ROM allows no reconfigurability of the VAD with the capacitor value in the CAP-ROM cell fixed. Yet, we will show that the VAD hit rate is independent of the training data, making an unconfigurable VAD bearable. Yang and Kim [22], [23] used a capacitor to reduce the power consumption of the digital ROM access, while we use the capacitor value itself as a memory for storing a weight without weight quantization. We propose the in-memory computing CAP-ROM to reuse the capacitor for both memory and computation, easing the readout circuit to reduce power and area.

After the Introduction, Section II describes the architecture of the proposed VAD and its circuit implementation. Section III analyzes how the binarized inputs and the CAP-ROM influence the VAD performance. Section IV shows the measurement results, and Section V concludes the article.

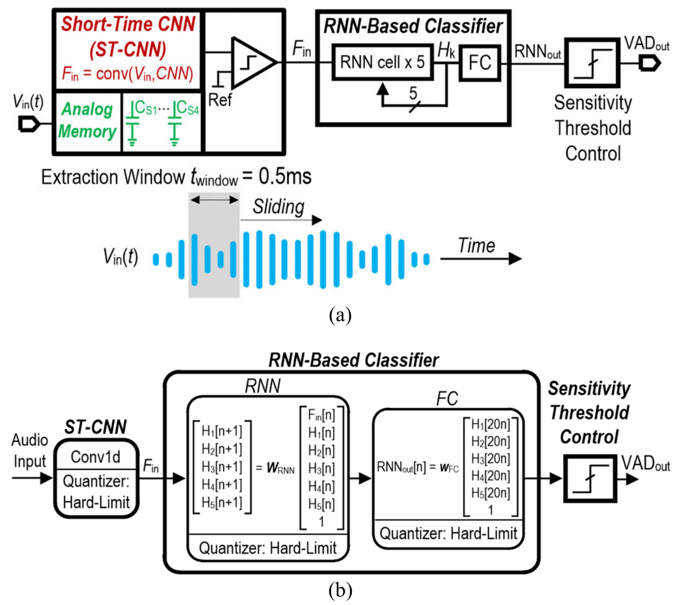

Fig. 2. (a) System architecture of the proposed VAD and (b) architecture of the VAD software model.

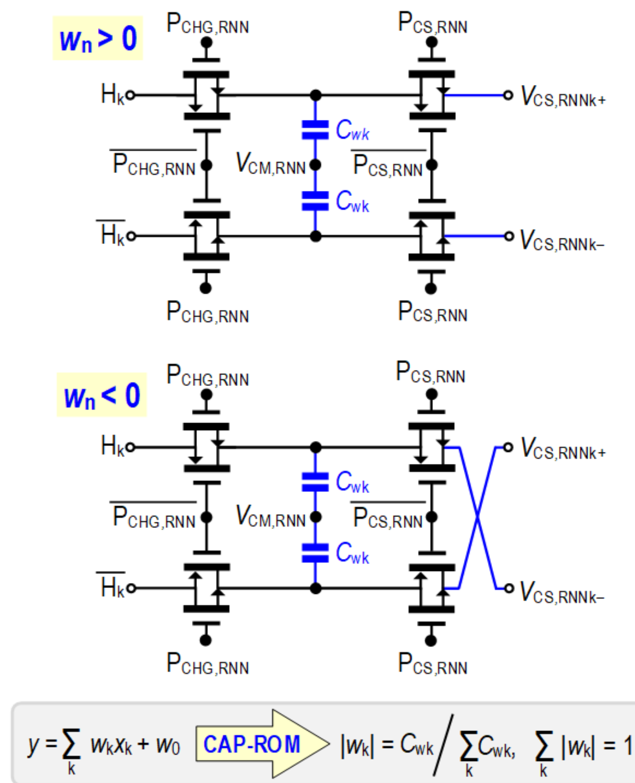

Fig. 3. Schematic of the CAP-ROM cell.

II. Proposed VAD Architecture

Fig. 2 shows the architecture and the software model of the proposed VAD, which samples the input voice at a rate of 8 kS/s. A 0.5-ms extraction window slides through the input signal and stores four samples in the analog memory. The ST-CNN is a 1-D CNN with only one kernel and a

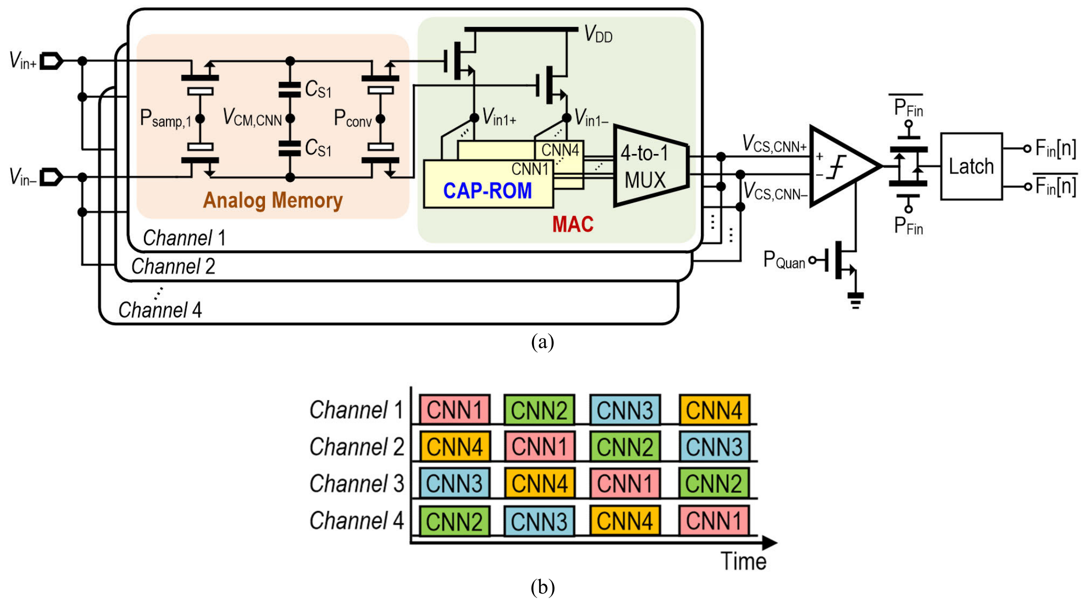

Fig. 4. (a) Circuit diagram of the ST-CNN and (b) timing diagram of the 4-to-1 MUX.

stride of 1. A comparator [24] quantizes the extracted feature (F_{in}) to one bit at a rate of 8-kb/s. The following single-layer RNN-based classifier processes the extracted feature, and a sensitivity threshold control [7] smooths out the output (RNN_{out}). In this work, the sensitivity threshold control outputs "1" if the number of "1" from the recent 10 RNN_{out} is at least 1. Otherwise, it outputs "0." As we choose this threshold value to be 1, the sensitivity threshold control does not increase the VAD's latency.

A. Capacitor-ROM

This work explores the capacitor as a non-volatile memory for charge-domain analog computation. We store the model's weight as a capacitance value embedded in the multiply-and-accumulate (MAC), being a CAP-ROM cell (Fig. 3). Each C_{wk} is a unit-capacitor array in the sparsity-aware computation (SAC) cell [7], where the corresponding weight determines the number of unit-capacitors joining the MAC operation in the SAC. Originally, we should store the weights of the TD-CNN in the ON-chip weight registers and load the weights for MAC operations; both are area-and-power hungry. Here, each weight determines the C_{wk} value in the CAP-ROM cell using the capacitor charge-sharing rule. There is no quantization since the capacitor can take continuous values. As the polarity of weight is fixed, we can cross-connect the output nodes if the weight is a negative value. A metal-oxide-metal (MOM) capacitor can provide a <0.4% mismatch, guaranteeing more than 8-bit weight resolution that is sufficient for different analog-computing applications. Thus, we use a capacitor to replace multiple bits of memory and the unit-capacitor array. It reduces the power for memory access and buffering of the unit-capacitor array. Moreover, it reduces the memory and the unit-capacitor array area as the utilization rate of the capacitor array is low (<50%). It also eliminates the weight preloading process before switching on the MAC. Even though the CAP-ROM has no reconfigurability, it suits specific tasks such as VAD or the feature extractor of a network that does not require weight updating. Also, it is possible to put the MOM capacitors in the CAP-ROM cells in a single layer to reduce the photomask-making cost while sustaining reconfigurability across designs by changing only one metal layer.

In this design, we store every weight of the ST-CNN and the RNN-based classifier in a CAP-ROM cell. The gain of the MAC is normalized to 1, similar to the TD-CNN. As C_{wk} in a CAP-ROM cell can take continuous values, we train the VAD model in floating-point precision and avoid weight quantization which usually degrades the accuracy of a neural network. With a 3-bit quantized model, the hit rate drops to 80%. We will provide an in-depth analysis of the CAP-ROM in Section III-B.

To limit the capacitor ratios of the CAP-ROM cells for better matching, we set a threshold of 0.2 during the training process. If the absolute value of weight is <0.2, we uplift the absolute value of the weight to 0.2 without degrading the VAD hit rate. The maximum weight of the model is 0.96 after training. Thus, the C_{wk} ranges from 2 to 9.6 fF. Reducing the threshold leads to a larger capacitor ratio and total capacitance. We choose the minimum capacitance of 2 fF limited by the parametric cell of the MOM capacitor. We can customize smaller capacitors, but it can worsen the capacitance matching and lower the gain of the MAC as the parasitic capacitance of the charge-sharing node $V_{\text{CS,RNNk}}$ is extracted to be 6 fF. A capacitor mismatch of less than 4% does not degrade the VAD hit rate, whereas the Monte-Carlo simulation shows that a MOM capacitor of 2 fF has a <0.3% mismatch.

B. Short-Time Convolutional Neural Network

Fig. 4(a) demonstrates the circuit schematic of the ST-CNN. We store every sample in the sampling capacitor (C_{S}) array in a cyclic manner. Each C_{S} buffers four CAP-ROM cells to output four products between an input sample and the four ST-CNN weights in the charge domain. A 4-to-1 multiplexer (MUX) in each channel picks the product to finish the MAC operation according to the timing diagram in Fig. 4(b).

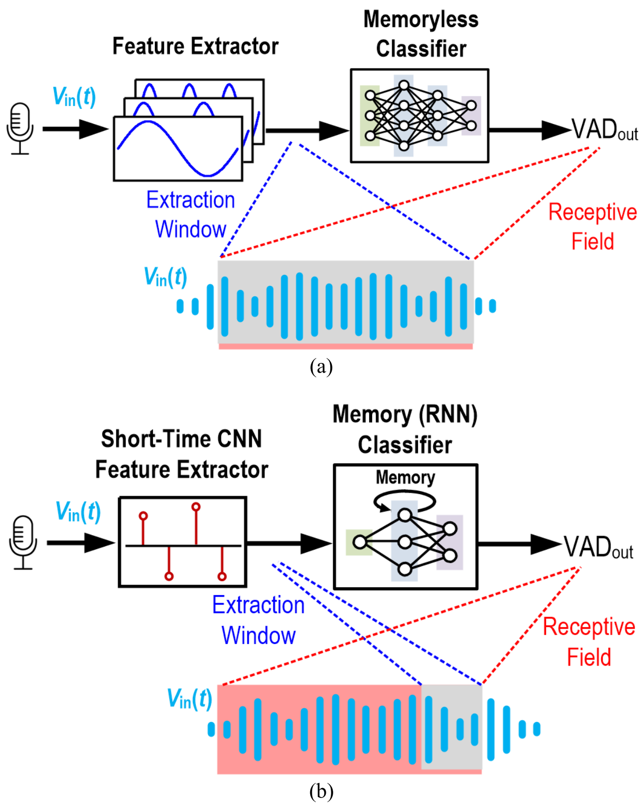

Fig. 5. VAD with a (a) memoryless classifier and (b) memory classifier.

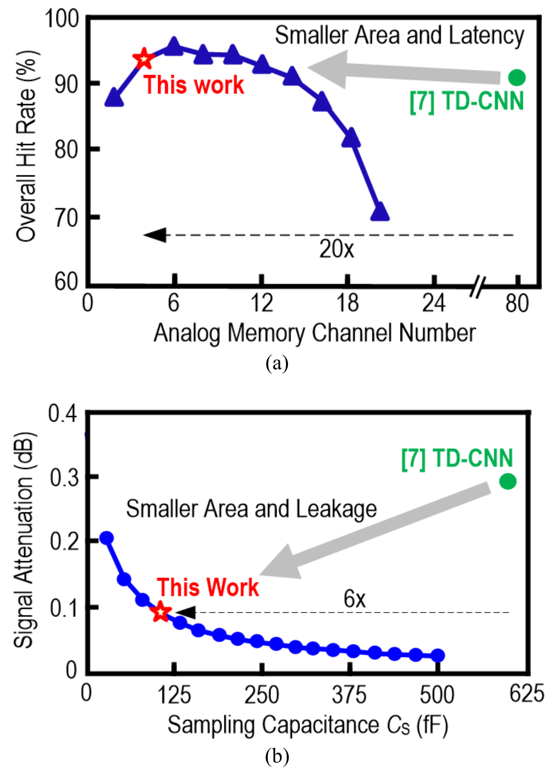

Fig. 6. Simulated (a) overall hit rate versus analog memory channel number and (b) signal attenuation over C_S.

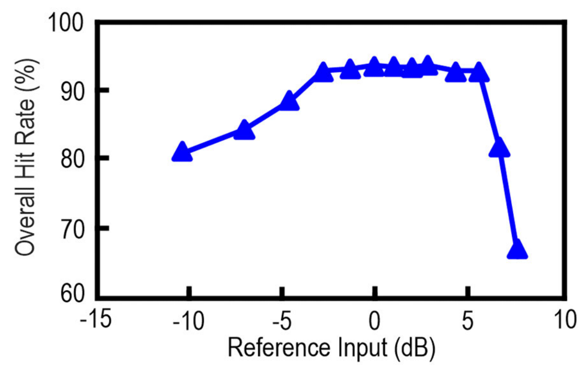

Fig. 7. Simulated overall hit rate versus relative input peak-to-peak voltage.

It prevents duplicating four sets of sampling capacitor arrays for area savings. The input signal convolves the ST-CNN weights, which are cyclically shifted in every sample. The thick oxide transistor reduces the signal leakage of C_S from 3.6 to 2.2 pA.

The receptive field length of a classifier corresponds to the lowest recognizable frequency. The extraction window is the same as the receptive field of a memoryless classifier [Fig. 5(a)]. It is typical to choose t_{window} at least 10 ms [5], [7] to extract features down to 100 Hz. We adopt a memory classifier to expand its receptive field from a shorter extraction window [Fig. 5(b)] as we should focus on the receptive field length. Fig. 6(a) plots the VAD's overall hit rate over the ST-CNN's analog memory channel number. A channel number of 4 corresponds to an extraction window of 0.5 ms at a sampling rate of 8 kS/s. A receptive field of 2.5 ms, extended by the RNN-based classifier, is sufficient to cover >400 Hz for sustaining the VAD hit rate where the dominant frequency of English is above 1 kHz [1]. The VAD hit rate decreases to <90% when the channel number increases to 16, as it makes the small model in this work spare parameters to recognize features down to 100 Hz which is out of the dominant frequency of English.

In comparison, the ST-CNN in this work has only one kernel with a channel number of 4 and a stride of 1, while the TD-CNN had 60 kernels with a channel number of 80 and a stride of 80. It is necessary to reduce the channel number of the ST-CNN with the CAP-ROM since the required number of CAP-ROM cells in the ST-CNN is the square of the channel number. Moreover, it corresponds to a shorter t_{window} in the ST-CNN (0.5 ms), so we can reduce the C_S for securing the signal leakage. Fig. 6(b) shows the signal attenuation of analog memory over C_S. We choose a C_S to be 100 fF to guarantee a <0.1-dB signal attenuation. The C_S in the TD-CNN was 600 fF which could only provide a <0.3-dB attenuation since the t_{window} of 10 ms was 20× longer than the ST-CNN. Reducing t_{window} can substantially reduce the size of the analog memory. The total C_S for the ST-CNN is only 400 fF, while the total C_S for the TD-CNN was 36.8 pF as part of the C_S was connected differentially [7].

Fig. 7 plots the simulated overall hit rate versus the relative input peak-to-peak voltage. The 0-dB point refers to an input peak-to-peak voltage of 700 mV. The overall hit rate sustains when the input signal is from −3 to 6 dB. The supply voltage clips the input signal if its peak-to-peak voltage is over 1 V. However, the signal clipping with a small clipping ratio does not degrade the VAD hit rate.

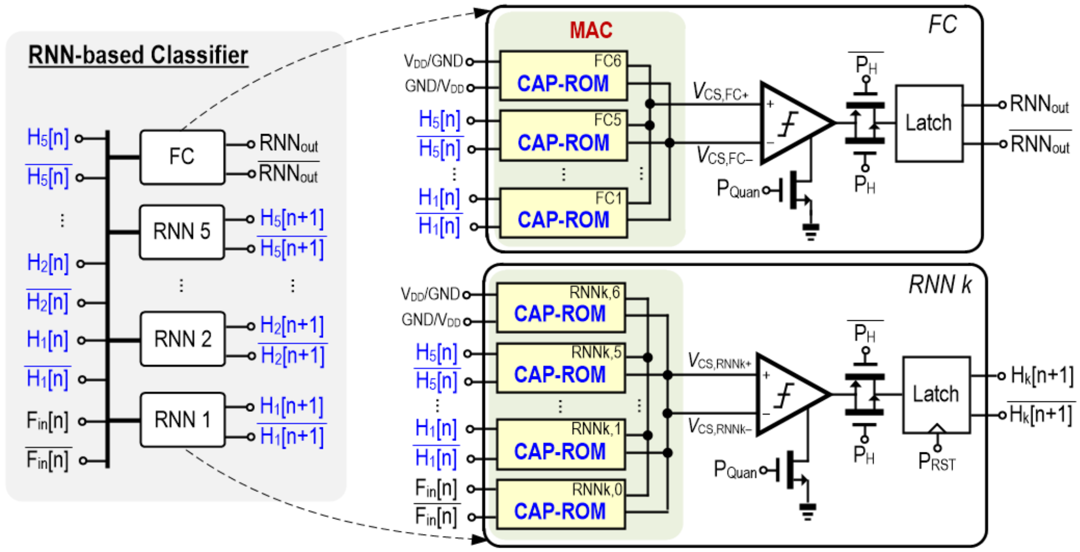

Fig. 8. Block diagram of the RNN-based classifier.

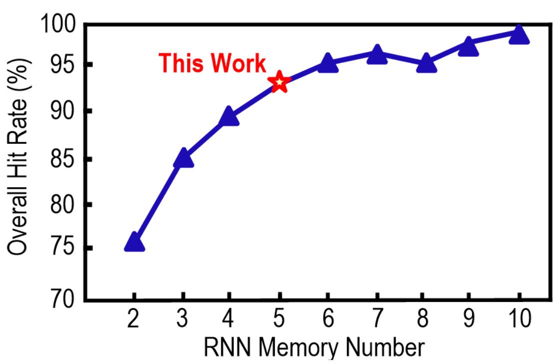

Fig. 9. Simulated overall hit rate versus the RNN memory number.

C. Recurrent Neural Network-Based Classifier

Fig. 8 illustrates the block diagram of the RNN-based classifier. It consists of five RNN cells for evaluating the memory states (H_{1-5}) and one fully connected (FC) cell for the VAD output (RNN_{out}). There are seven and six CAP-ROM cells in the RNN and FC cells, respectively, where the last cells of them ($\text{RNN}_{\text{k},6}$ and FC_6) represent the bias term of the MAC. An RNN cell has six 1-bit inputs, including F_{in} and five memory states. The FC cell only has the five memory states as its inputs.

Every RNN cell updates a memory state parallelly at a rate of 8 kb/s, while the FC cell only computes RNN_{out} every 20 cycles, which makes the classification rate of the VAD to be 400 Hz and a latency of 2.5 ms. We reset the memory states after the FC cell outputs a classification. Unlike the ST-CNN, the RNN and FC cells do not include a MUX to select CAP-ROM cells for MAC operation since all the inputs of the RNN-based classifier are in the same sample. Fig. 9 shows the simulated overall hit rate versus the RNN memory number. We choose the memory number to be 5 for a 94% hit rate. We can increase the memory number for a higher hit rate while it increases the power and the area of the classifier. The RNN-based classifier reduces the model size by having more weight reuse. The proposed VAD has only 45 parameters which are 164.8× less than the TD-CNN. This work has 1572 operations in 2.5 ms for one classification, while the TD-CNN has 14 832 operations in 10 ms. Thus, the weights in this work and the TD-CNN are reused 35 and two times, respectively.

III. Performance Analysis

A. Discretized Output Distribution

The TD-CNN had a dual-peak output distribution due to the sparsified quantization (SQ) and a kernel size of 80, which is smaller than the traditional MAC. The ST-CNN has a continuous normal distribution similar to a regular MAC operation. Yet, the RNN and FC cells have only six and five binary inputs, respectively, which makes the RNN/FC cell has at most 64/32 possible output values. Thus, the RNN and FC cells have discretized output distributions (Fig. 10), providing larger margins to the following comparators to desensitize their offset and circuit noise. Fig. 11 shows the Monte-Carlo simulation results of the VAD hit rate when considering the comparator's offset. The overall hit rate starts to degrade when the standard deviation of the offset is larger than 5 mV, while the offset of the VAD under circuit-level simulation is 3.4 mV. The simulated rms voltage from charge injection and clock feedthrough is 1.9 mV, and the simulated input referred noise of the comparator is 0.1 mV, which guarantees the VAD's robustness.

B. Necessity of the CAP-ROM

Even though the CAP-ROM provides different levels of advantages, it is tempting to apply the SAC + SQ [7] to this work. Fig. 12 compares the simulated hit rate over the weight resolution with SAC + SQ and traditional uniform quantization. We observe that the SAC + SQ is still superior to the uniform quantization as the SAC + SQ can increase the resolution of the weight quantization [7]. However, the SAQ + SQ's improvement is limited in this tiny model where the kernel sizes of the RNN and FC cells are six and five, respectively. Thus, it requires a 7-bit SAC + SQ weight

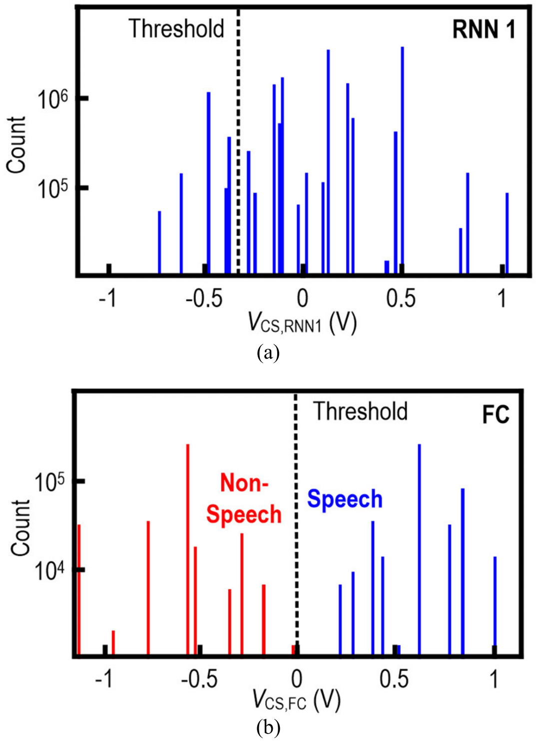

Fig. 10. Output distribution of (a) RNN_1. (b) FC cell.

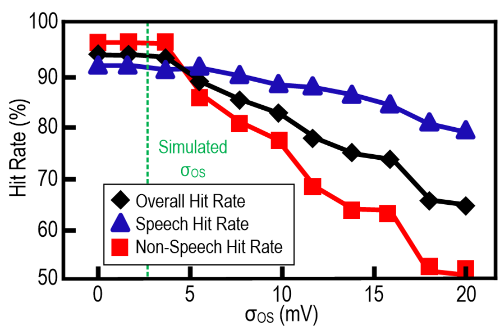

Fig. 11. Simulated overall hit rate versus comparator's offset.

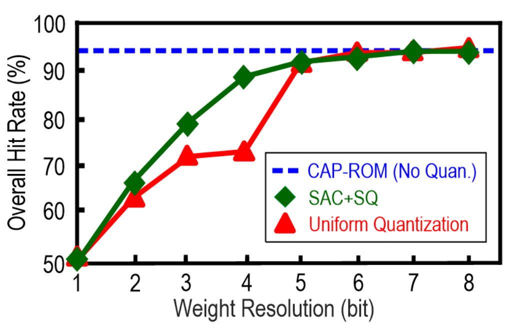

Fig. 12. Simulated hit rate versus weight resolution for different quantization schemes.

quantization to achieve a 94% hit rate as the CAP-ROM without any quantization. Therefore, we need 315-bit memory for storing the weights. Also, as a 7-bit parameter with SAC + SQ requires 126 unit-capacitors, we need 504 and 756 unit-capacitors for the ST-CNN and RNN-based classifier, respectively, where the RNN and FC cells can share the latter

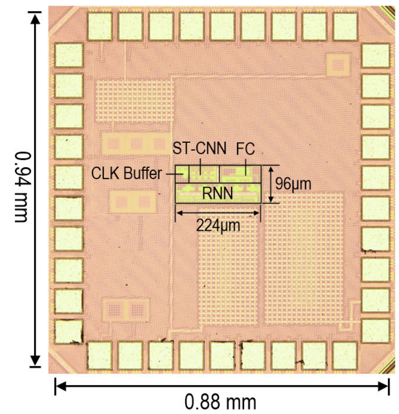

Fig. 13. Chip photograph of the VAD.

in a time-interleaved manner. Comparatively, the required number of unit-capacitors in the TD-CNN was 480 with 3-bit of 80 parameters for each kernel. The SAC + SQ can decrease the required weight resolution to 3 bits when the kernel size is 80. However, the RNN-based classifier has small kernel sizes of five or six, which weakens the improvement of the SAC + SQ, making the required number of unit-capacitors for both models close. It means the effort of compressing the model turns to a higher weight resolution. Assuming the computation cell using the SAC + SQ has a unit-capacitor value of 2 fF, the total required capacitor array value is 2.52 pF, 6× larger than the analog memory in the ST-CNN. Comparatively, the total C_{wk} in the CAP-ROM cells is only 195 fF in this work.

In summary, if we apply the SAC + SQ to this work, we need to add 7 bits of weight memory and replace each C_{wk} with 126 unit-capacitors. Regarding power, a CAP-ROM cell with C_{wk} of 2 fF consumes 2.5 pW at an 8 kHz clock at a 1-V supply, while the leakage power of a 6T-SRAM cell with hvt transistors is 10 pW. Thus, adding 7 bits of SRAM for each CAP-ROM will increase the power by 28× without mentioning the power consumed by a larger unit-capacitor array. In terms of area, assume the density of a 6T-SRAM cell in a standard 65-nm CMOS process is 0.5 $\mu\mathrm{m}^2$/bit [25]. The area for 315-bit 6T-SRAM is 157.5 $\mu\mathrm{m}^2$. The density of a MOM capacitor is 1 $\mu\mathrm{m}^2$/fF, so the total unit-capacitor of 2.52 pF consumes 2520 $\mu\mathrm{m}^2$ area while the total C_{wk} in the CAP-ROM cells only consumes 195 $\mu\mathrm{m}^2$. Thus, applying the CAP-ROM to a compressed model is necessary for this work rather than using the SAC + SQ.

IV. Experimental Results

Fabricated in a 65-nm CMOS process, the proposed VAD occupies a chip area of 0.94 × 0.88 mm^2, including the pads (Fig. 13). It consumes 47 nW at a 1-V supply. Fig. 14 summarizes the power and active area breakdown. The active area of the VAD is 0.022 mm^2 whereas the remaining area mainly includes decoupling capacitors and dummy fill as the chip is pad limited. Also, we eliminate the power-and-area-hungry weight register, although there is no reconfigurability of the VAD. Fig. 15 demonstrates the test setup of the proposed VAD. An Agilent Signal Generator E4438C generates the

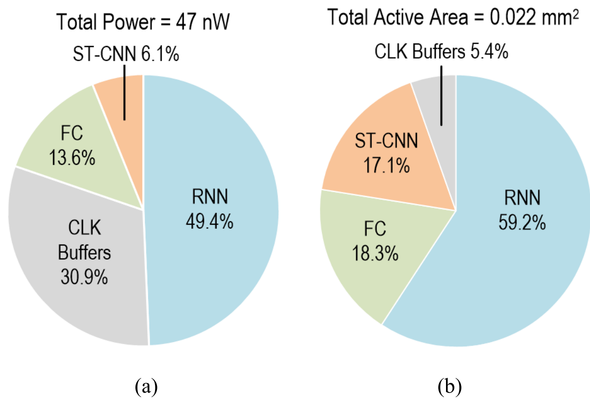

Fig. 14. Measured: (a) power and (b) active area breakdowns of the VAD.

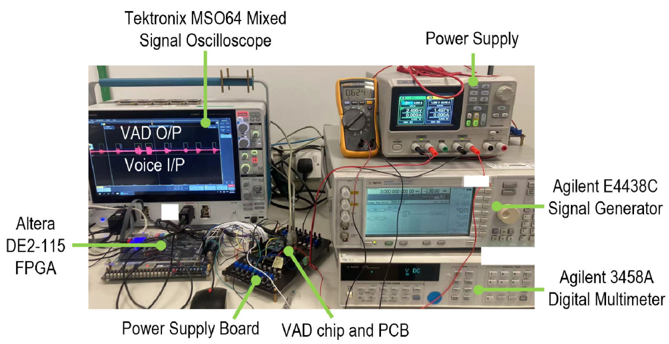

Fig. 15. Test setup of the VAD.

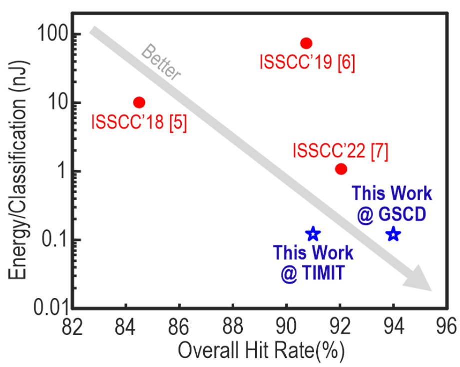

Fig. 16. Measured energy/classification versus overall hit rate.

input signal. An Altera DE2-115 FPGA generates the required clock signals whose frequencies are not more than 500 kHz. A Tektronix MSO64 oscilloscope shows the input and output of the VAD and stores them for hit rate analysis.

As the VAD's parameters are fixed due to the CAP-ROM, we trained our model with the Google speech command dataset (GSCD) with the babble noise from the NOISEX-92 dataset of a 10-dB signal-to-noise ratio (SNR). Yet, we test the VAD with both the GSCD and the Texas Instruments/Massachusetts Institute of Technology (TIMIT) dataset, both with the babble noise of different SNRs, to show that the VAD's hit rate does not strongly depend on the training set. The TIMIT dataset was recorded in a better-controlled environment [26], so we expect a VAD model trained with the GSCD, recorded with different microphones by different users, can sustain its hit rate on the TIMIT dataset.

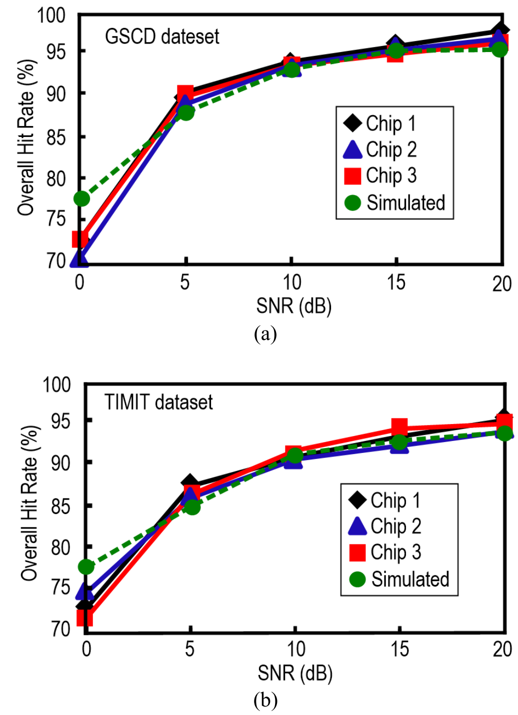

Fig. 17. Multichip measured and simulated overall hit rate of the proposed VAD on the (a) GSCD. (b) TIMIT datasets.

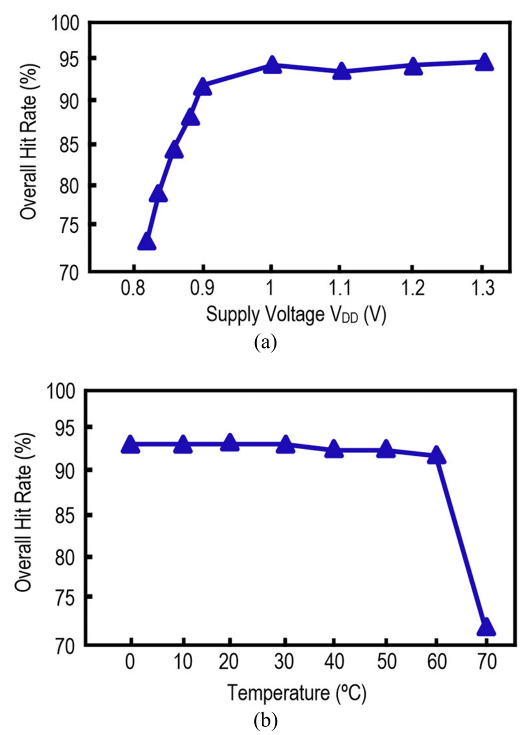

Fig. 18. Measured overall hit rate versus (a) supply voltage and (b) temperature.

The measured overall hit rate on the GSCD and TIMIT datasets at a 10-dB SNR is 94% and 91%, respectively. The classification rate of the proposed VAD is 400 Hz with an energy per classification of 0.12 nJ. Fig. 16 compares the measured energy per classification and the overall hit rate with the recent VADs. The two markers representing this work are at the bottom right of the figure. Fig. 17 exhibits the simulated

TABLE I
PERFORMANCE COMPARISON OF THE FEATURE EXTRACTOR

Feature Extractor	This Work	JSSC'22 [7]	JSSC'19 [6]	JSSC'18 [5]
Feature Extraction Topology	**ST-CNN Feature Extraction**	TD-CNN Feature Extraction	Time-Interleaved-Mixer-based Frequency Ext.	Analog-to-Event Filter Bank
Programmable Feature Extractor	No	Yes (TD-CNN + Sparsity SC)	Yes (b-DCT Sequence)	No
Feature rate (kb/s)	**8**	6	16	14.4
Window Length (ms)	**0.5**	10	16	25
Feature Type	**Binary**	Binary Adaptive	Digital Frequency	Event-based Frequency
Supply Voltage (V)	**1**	0.9, 0.6	3.6, 1.4 & 0.6	0.6
Power Consumption (nW)	**2.9**	73*	60	380
Area (mm^2)	**0.0037**	0.055	0.56	1.6

** power consumed by the clock buffers for the sampling, charge-sharing buffers and LNA*

TABLE II
PERFORMANCE COMPARISON OF THE VAD

Voice Activity Detector	This Work	JSSC'22 [7]	JSSC'19 [6]	JSSC'18 [5]	JSSC'21 [8]
Classifier	**RNN with CAP-ROM**	TD-CNN + BNN	Neural Network	BNN	Analog SNR Decision Rule
Reconfigurable	No	Yes	Yes	Yes	Yes
Classification Rate (Hz)	**400**	100	2	100	31.25
Parameters (Resolution)	**45 (No Quantization)**	7.4k (3 & 1-bit)	1.6k-11.9k (4-bit)	4k (1-bit)	N/A
Dataset	GSCD + NOISEX-92	TIMIT + NOISEX-92	LibiSpeech + NOISEX-92	AURORA4 + DEMAND	Custom
Speech Hit Rate/ Non-Speech Hit Rate	92%/96% @ 10dB SNR	90.1%/94% @ 10dB SNR	91.5%/90% @ 10dB SNR	84%/85% @ 10 dB SNR	Not Comparable
Power Consumption (nW)	**47#**	108*	142	1000	760
Energy/classification (nJ)	**0.12**	1.08	73	10	24.32
Latency (ms)	**2.5**	50	512	30	N/A
Chip Area (mm^2)	**0.8**	0.8	17.6^	2.5	0.14 (Active)
Supply Voltage (V)	1	0.9 & 0.6	3.6, 1.4 & 0.6	0.6	1.2
Technology	65 nm CMOS	28 nm CMOS	180 nm CMOS	180 nm CMOS	180 nm CMOS

 # No additional power required for memory access *^ Area including an audio compressor and a processor*
** Excluding weight memory power*

and measured overall hit rate versus different SNRs for three chips. The overall hit rate is higher than 92% and 90% on the GSCD and TIMIT dataset for at least a 10-dB SNR. As we cannot configure the parameters of the proposed VAD, we also measured the hit rate with different supply voltages and temperatures to verify its robustness (Fig. 18). A supply voltage from 0.9 to 1.3 V or a temperature from 0 °C to 60 °C does not influence the VAD hit rate. Based on the further simulation, we investigate that a low supply voltage or a high temperature only affects the bias voltage of the source follower buffer in the ST-CNN. Thus, we only need to adjust the reference voltage of the following comparator with a simple temperature calibration. Also, a lower supply voltage increases the input signal clipping. It is possible to improve the VAD hit rate at low SNR by using mixed-SNR training data while it lowers the hit rate at high SNR.

We benchmark the performance of the feature extractor with the state-of-the-art in Table I. The ST-CNN reduces the window length to 0.5 ms due to the RNN-based classifier. There is less signal attenuation, and the analog memory is smaller than [7]. Thus, it consumes 2.9 nW, at least 20× less than the others, and its area is 0.0037 mm^2, which is 15× smaller than the TD-CNN. We compare the performance of the VADs in Table II. Although the ST-CNN has only one kernel, its feature rate is similar to [7], where the RNN-based classifier combines the recent features for lower frequencies' information. The RNN-based classifier also reduces its number of parameters, significantly reducing the VAD model size to 45 parameters, at least 35× less than the others. As the ST-CNN has a short window length, the latency of the VAD reduces to 2.5 ms. The achieved energy per classification is 0.12 nJ which is 9× less than [7] as the operation per classification is also reduced by 9.4×. The energy efficiency of the proposed VAD is 13.38 TOPs/W.

V. CONCLUSION

The article introduced a VAD using an ST-CNN and an RNN-based classifier with CAP-ROM as the weight storage. The RNN-based classifier shortens the extraction window, making the feature extractor the ST-CNN. A shorter extraction window reduces the size of the analog memory and signal leakage, improving the computational precision and reducing the VAD latency. Storing the weight in the CAP-ROM can eliminate the ON-chip volatile memory and the related power for memory access. Moreover, using the capacitor as a non-volatile memory can represent high-resolution weight value, avoiding weight quantization which usually degrades the VAD hit rate. Although the CAP-ROM has no reconfigurability, the measurement results show that the VAD hit rate does not strongly depend on the training data. Also, the required number of CAP-ROM cells is around the square of the kernel size. Thus, the CAP-ROM is suitable to apply for a small model. The RNN-based classifier significantly reduces the model size of the VAD. We also realize that the SAC + SQ will lose its effectiveness for a small kernel size, rendering the CAP-ROM well-suitable for the proposed VAD.

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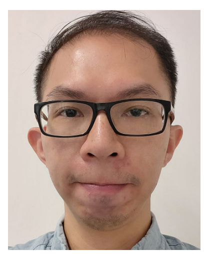

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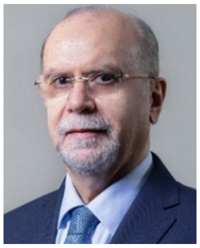

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